

ActInf Textbook Group ~ Cohort 6 ~ Session 18 (Chapter 7, Part 3 / Applying

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okay we're in cohort sixo question mark

And discussing some applying active

inference questions and Jeff has turned

us to the inspiration topic

so where were we Jeff or

continue yeah so um several of us Dean

tickles and Anna magdalena herado and um

um Hector Manriquez we've been going

back and forth for for about two weeks

on on what inspiration means uh through

the lens of active

inference and and for me inspiration and

I think this is a really good um

definition and I think Dean offered this

up and it was um what happens when you

have

a new

new

model or a new

artifact that has

no through line no uh

contiguity

for how you receive how

you

arrived at

that

specific phenomena or or or experience

or or bit of

information right it is it is that that

is seemingly without

cause

Okay lot of ways we can model it just

are there any more

General comments or or definitions or or

thoughts and then otherwise we're g to

kind of walk this pseudo chapter
6 path for H how we would move towards
structuring
it you're basically asking like how does
like Insight arise into like the shape
of some like object you're looking at or
some insight you're looking
at is that what that last question
Zach yeah this uh this reminds me of
really core active entrance right
whenever you're talking about like if
you just have a whole bunch of particles
you can differentiate the states of the
particles are in by the states that seem
to have this kind of sub thing going on
and the maintenance of the sub thing and
this thing want
by virtue of existing very core
definition of active inference right or
I'm sorry the uh I'm forgetting the word
of it though it's a very important word
the idea that because it exists we can
talk about the Dynamics of it
existing I'm sorry I can rephrase that
if you I did not that very conditioning
conditioning modeling of something on
that thing being like existing or
measured yes absolutely
and so it's going to maintain its own
state
but the question I have is could you
guys describe the no three line or
continuity because by saying that it is
a thing aren't we already talking about
it has a thingness about it we are
already talking about some states where
it is instead of some states where it is
not yeah like even a really in inspired

new line of poetry
it's like it it may have a chain of
causation so then if we were if we were
um tasked to make an active inference
model um and in that first question
chapter six like what system are we
modeling it's like what what question
are we trying to
explore are we trying to reify our
definition or
observation that inspiration is that
that arrives at with without uh uh cause
or are we we looking to quantify like
some index of inspiration where on the
lower end there's like complete
continuity and then on the upper end
there's less

continuity here's one

Boulder

paper with the kind
of more on the the Insight

[Music]

um

okay so from chapter six we have those
four stages of the
question let's just answer the first
question and then we'll look over to to
a few other

um kind of uh tools for modeling but
what what system are we going to model
like can we pick some specific
inspiration setting just to make it
clearer

a song comes to an artist in a
dream

okay here I'll I'll I'm going to pop
over to this other um active block
frence Koda just

because um it it complements the chapter
six so in the textbook chapter
six gives these four
steps um that suffice to design active
inference generative model so big
picture the first part is like system of
Interest identification systems
Engineering Systems modeling then you
get into the kind of chapter 4 chapter
seven and eight questions like is it
going to be a discrete time or
continuous time what kinds of State
spaces are the variables and then
question three and four are about
setting up basically the agents and the
niche and it's this is dealt with at a
pretty overview level in the
textbook and um depending on how
how complicated the setting is that this
can be like very straightforward like
thermostat and the room or this might be
really uh sophisticated like having a
bunch of Nest mates in in a colony but
here here's just a few um compliments to
those
lists here's from the blockfront
side
um the cognitive modeling side first off
the settings
many discussions over the years and and
sub discussions and things like that
that that people have brought up but
just focusing on the
recipe um AI um
this kind of just gives a little bit of
a of a different flavor which is first
take take a structural approach to the
project like it asks a few more

background questions about what what is
being addressed and not then this was
like some some of the earliest um memes
with yakob which was
like we're we map building with these
cognitive models and then we're playing
with this like der
because you could derive something is
kind of like to to squeeze out but then
also it means to take the derivative and
then there's different notations for the
derivative like sometimes it's a DOT or
it's a prime or it's like f' of x so
then if we derive a model but then
there's different notations for the
model and then the table that I'll copy
back into our
um textbook
group
this I'll just take it from the
ant model
okay here are active inference ontology
terms I'll delete these
columns and their um notation so we see
 a b CDE e
 π
aocs and then here we'll we'll do artist
generative
model once we fill out
these
cells for like if if we're going to use
a discrete Time model um then we'll
basically have all the variables we
need but this is just a way to use the
ontology to scaffold
making a generative
model because basically you just fill
out what would these things be for that

situation and then already just manually
or with a llm there's enough information
to get you to the
PDP

okay what should the observations be
not the components of the song I
suppose yeah this also this is
like part of that beginning um
going from the qualitative and starting
to give some structure it's
like is it going to be just a sound
model and then H how many layers to
go into
detail but that's kind of that that
journey is to take the the um pre
structured
situation like just situation as it is
or or or or um uh not exp explicitly
categorized situation and then be like
what are we going to pull out as
observations for
whom so it's like like will there be a
bodily awareness will there be a time
awareness if if we're going to ask a
question about time then I mean we may
need that to be explicitly included or
but maybe it's not in the
observations let's start with just music
sound and we can figure out what types
okay what hidden states are
being
inferred so what are what are
um unobserved but hidden causal factors
that actually exist or are proposed to
exist
if there's lyrics then you have to
suppose the understanding of the
language same with instruments right you

have to have a model of of which sounds
or at least how you can instrument the
sounds that
you yeah how you can instrument the
sounds
there's also an interesting piece there
with with this um if if we're going to
come back to the time perception because
music has a Time
rate but like a architectural Vision or
something that might be static or so
then but the
song I mean does the does the um artist
hear the song at Onex
speed or do they hear it like faster or
slower and then and then what if it's um
not possible to be played or do they get
it partially and then fill the rest of
it in when they when they wake
up yeah true underlying song like I it
like it sounded like they were
underwater but I heard
something okay
let's just
also just to be clear with the letters
and I'll just bring it over
okay so we have some observations coming
in at the very least
music possibly some time
observations or some
rhythm in the
S we
have two two kinds of s's
States related to the music and then
States related to the flow of
time
okay what does that give for the a
matrix

then the the simplest answer is a is
mapping between s and O what whatever
those are so that could be pure intra
domain so that would be
like sound o maps to sound
s time o maps to Time
s so in that situation there' be like
two separate
s um
variables they could be different shapes
and then each of them has an a leading
to separate o
variables or there can be more of a
domain Fusion like um
single um
s with mapping
to sound o and sound and time o
me this is like why it's just like get
it down and like kind of iterate over it
it's like oh because like what is what
is the really the
time observations like well is is that
an internal observation is that a
metronome is that a
breathing so it's it's like we can kind
of run back over and check for for a
consistency but basically a is going to
do whatever intermediation work is
required to get from o to s
but this is the sense making partially
observable element that's going to go
from whatever you're saying is data as
measured to variables as
proposed just thinking of like another
ingredient like sometimes as a musician
you can like feel the song in terms of
like how you hear it more like through
your spine or like energetically but

then you you might also like that might
get mapped to how it feels like to play
the song you might have some kind of
intuition of like the fretboard or
whatever it
is yeah that's good good idea like using
um kind of like
Intermodal her istics like um this cake
is going to taste good because I just
touched it and it's
bouncy so then there could be in that
situation that would be more like the
domain Fusion for a because you'd have
like the Latent state would be like
quality of the cake and then it'd be
like um here's how the color uh profile
is associated with it and then here and
then here's how the um touch here's how
the springiness is associated with that
latent variable of cake
quality but then in the dark room you
still could use the touch modality to
assess the cake
quality
okay let's move P_i up so what actions
are being
modeled as being
taken actions are going to
intervene they're going to pick out a
slice of
B and then the chosen slice of B is
going to be used to control the time
evolution of
s so again kind of an interesting case
like what are the
actions in this situation where um the
inspiration is putting on a show for the
artist what action selection are we

looking at for them

so I can get grounding we saying like
the action is like basically like as the
dreamer like as the person's like waking
up from the dream and trying to like
grasp onto the song and and solidify it
or is this like a different kind of
question we can make this really like
non-physical right because it's coming
from a dream therefore we can say well
could we say that like one of the
actions is changing our model of the
world as in now I want to make this song
before I did not want to but then I had
the dream which is a perception and the
action I take is the desire to make this
thing in

reality okay yeah let's let's get some
of those so let's just say that one one
kind of action would be if s
equals

motivation to make song

would

be

um so so through time there's this
motivation to make some song and then
there's B that simple two two options
for B either keep your motivation where
it was or increase your motivation would
be choosing to increase
motivation obviously other ways to model
that too here if s

um Martin brought

up this like waking

up there's modeling of attention as
action that's some of the work with um
lar SED

Smith where

internal action policies covert action
policies are modeled as attention so
this is like s would be what am I paying
attention to and then P_i is a policy on
what is being attended
to but it's but but kind of using this
skeleton like our um recipe
outline it's
like
whatever whatever s
is there's going to be some s's that
don't have direct policy control those
are called uncontrollable States like in
pmdp um and then there are the s that
are
controllable which means that their bees
their their their unfolding is
controlled by bees that have some
policy efficacy
but then this this this dream um setting
kind of um
stretches the case because it's like is
this a lucid dream where the person
is
seeking to
um you know is is it really them from
behind the veil inspiring
thems but then obscuring that it was
them
but then this goes back to that first
question of chapter six which is like
which system is are we modeling and um
inactive inference just like any
anywhere else people have done systems
modeling that's not necessarily trivial
because once you start tracing the
physical and the informational exchanges
of a system you can you know you you tug

on one part of the world and everything else comes along with it so that that does become the question like which modalities to restrict to on the other hand a great strength of active inference is that if the models are constructed appropriately at least they'll be interoperable like you can say we're making just the dream auditory track and then just the dream proprioception track and then you could test those two models separately and then you could more easily also fuse those models see if there was like um synergistic outcomes so there's still a lot of possible modeling but at least what you make maybe more interoperable let's see what other s could be get an idea of like the framing here maybe it's a question but um the π is like some kind of like action that changes the state but does that also like create the flow of time in a sense that just how States change over time yeah great question just the the narrower answer is π is a probability distribution over policies policies are the affordances that can be taken in a given moment over the given time Horizon so if there's no planning then π are just the actions that can be taken in the next moment which are the affordances like up down left right if there is planning then π starts getting bigger like exponentially bigger with the time Horizon like two time steps um

if there was four affordances there'd be
16 options in pi so Pi is a probability
distribution it sums to one because some
action is taken and then depending on
which of those um indexed policies are
selected that's which B transition
operator is chosen and then that's what
controls how s at T minus one goes to S
of t Okay so but then when it comes to
modeling it's like um and again time
time perception is another another fun
and interesting case but it's like
whatever what are it doesn't reach in
directly and change the temperature of
the room it's like temperature of the
room at the previous time and then you
decide turn on the air conditioner or
not and then if you choose turn on the
air condition conditioner then you got
the prior t uh temperature turn on the
air conditioner and that results in the
next
temperature so it's not that that Pi
directly like goes in and
changes the temperature in the room but
it it acts through this
B um per yeah appreciate that
where and and there have been a
few time perception active inference
papers I'll try to
find yeah we're definitely playing with
the weird system because time perception
and dream
changes okay
here's Bogata and
Jaa I I forget which number we can look
but it was a live stream
previously so

here here they're using the
haerian phenomenology ontology so
retention is what is retained or
sedimented from the past prot tension is
what is anticipated into the
future so this is kind of like the
thickness of the presence where like
what
was what was uh
anticipated moves through the the
present into the um and then be parts of
it becomes
retained
here okay this might help us
see here okay font looks a little bit
odd but this is this is State
s
o this is
basically this the the table
don't know where the table were but this
is basically what the table is and this
like this is what applying active
inference looks like in this setting
where we're doing the ABCDE e
o it's about defining what those
variables look
like for the given system or a question
of Interest
is it just me or does that look a lot
like relativity except the gamma term
has been replaced by a probability
distribution which which part uh mostly
that first graph that you showed where
uh there was that kind of diagonal set
of squares that's what really set it off
for me uh and talking about uh the
object time versus the model time uh
using to and T that was another thing

that made it really seem similar that
one right there exactly
so what what would this be like in um
relativity that would be uh changing
reference frames which would mean that
like they are perceiving space and time
differently than a different Observer
would which is to say that like one
person at point B would look at
something and say oh uh maybe Point uh
I'm sorry a top left d uh would look at
something and be like oh I see two
events in two different points in space
they happen at time one and time two but
at person's point AF of to the far left
would look at those same two events and
say no they didn't happen at time one
and time two they happen at time three
and time four they would happen at
relatively different times because their
experience of time would be slightly
different oh yes I see this like or it
could be like here's here's um message
one two and three person A has gotten
none of them person B has like yeah I
got message one but but I'm just getting
message to now but then for this person
they were closest to this I don't know
if it's exactly how it's laid out here
but it's like already for them the
messages are in the past but for someone
else those messages haven't even
happened yet yes exactly the time is
relative okay we we'll see what we can
bring back into the model but also looks
kind of like a
fretboard like a diagonal line on a
fretboard for a base

well it totally
does we we can make him um song
compression okay that I mean was that
just was that just an example of
inspiration
there it might have been I don't
know it looks like
it how much how and also this is kind of
related to heck
and Michael Walker's work on the side by
side in the zone of of bounded
surprisal which is like there's um the
question of the the insight to One agent
and then also there's the question of
like whether it can be
observed and whether that can be
transmitted like the the just to make it
more concrete there studying these
flaked Rock
artifacts that that um like sort of
human ancestors made and then they're
always like popping up but then they go
away but like it will be it will have
been made a certain way but then it's
not transmitted how it was
made so just because something can can
can be seen by one doesn't make it seen
by another
let's just see if there's any other
pieces to this paper then we'll return
to where we
were I mean the this
is there's obviously a lot to it here
and they're they're going into the
they're kind of Bridging the H
seran
phenomenology with the future present
and

past with the active
style but that's kind of the meta rest
recipe which is combine the actim
fonty the the letters and the functions
and the methods that we know with some
other

domain time Consciousness is kind of a a
particularly there's it it it has

because the the

um the measurements

come through another

modality there's not like approximate

sense like you could have a a a

vibrating clock or a metronome or

breathing but then the time inference is

something that's added on to some other

modality whereas in sound it's like

sound observation and then you're

observing a sound latent state

let's just pick one kind of policy to

model if s were true song

Tempo Pi would be speed

up stay

same let's just say that it's a elcid

dreamer with with this limited kind of

control over the music that they're

hearing they can they can change the

tempo so if if Pi has three

elements for slow down stay the same or

speed up up so this would just be like a

simple Vector with

three um cells if it were uh uniform

it'd be like $\frac{1}{3}$ $\frac{1}{3}$

$\frac{1}{3}$

see these are preferences so does the

person have a preference for the

observations is that kind of like if the

song seems Salient to them they'll try

to like remember it or something like
they just kind of reference of what the
song feels

like yeah I mean we it would have to be
sketched out but what this this is going
to

go this is going to load on pragmatic
value

C is basically is is 100% where
pragmatic value gets
specified if C is

flat then there's no pragmatic value in
the generative model you only have
epistemic

value so it's kind of interesting like
um is is it is the song recognized as
valuable because it's

what they

prefer or is it recognized as as
insightful because it is recognized as
novel something could be both novel and
preferred however if if you recognized
to the extent that you recognize it as
part of your prior

preference that would not be novel
or if there's like notice that there's
some kind of like Beauty to it and the
sense of like truth beauty and goodness
like it's maybe some kind of meta
structure that has some geometry to
it yeah I

mean this is why we're um why in the
textbook it's like the
teamas and and the simpler cases
because once we're dealing with a kind
of um mental
kernel that that has these um that for
any input is going to start exploring

like good true beautiful structured
Etc then it's like um it kind of makes
the sensory inputs just a small piece of
the picture whereas in the textbook
examples it's a very kind of
closed form model where you can trace
exactly how the observations relate to
like changes in the model's
Behavior so D this is the prior on
hidden
state so if we're going with s is the
song
Tempo let's just say prior on initial
 S so this is kind of like
interesting um maybe even the song Tempo
could relate to the time and the Rhythm
so it's like if we're hearing
sounds we hear one
beat so but we haven't heard a second
beat yet so we can't yet estimate how um
how fast the song is so all you have is
the prior on the
song that's that would be all right when
you're right
here right yeah e this this is the
Habit that habit this is prior on
policy um it it's it's not it's it's
right here e um so this is this could be
uniform or not but basically just your
your prior on policy and then a is is
the action but we we it's the it's what
policy
selects okay then
what about the generative process again
this is a very interesting question
where it can go like back and forth
between sketching out a little bit about
the agent then going back to the niche

like so what is the niche
of this artist that we've been sketching
out
what you mean by Niche like a musical
Niche like the genres they they like or
something
um I think it it could be um I'm just
going to go to the the paper with the
attention
but like just just generically the
observations that the artist is getting
are coming from the niche and then the
actions that the artist is emitting are
being received by the
niche so it's just that's when you plug
in the agent model into the
niche so that can be um like in the case
of sensory data and action like bodily
overt
action that um is sometimes related to
the physical Niche like it's getting
thermometer observations from the room
and then it's sending control signals to
the air
conditioner but then
for example in this is the model where
policy is is
um here like here this let's look at
this top level um
attention let's look at okay just the
two- Lev model
OKAY the blue is the proximate sensory
so this could be like visual data from
the
ey and then here is on the the orange
levels nested model it has the same
structure as as this as the
PDP um except come in sa and then

instead of

Sao instead of like um what shape is or what color am I seeing and then what are the raw pixel data coming in that's the primary approximate visual layer but now here the SA but then the observation that this higher order agent basically this metacognitive agent is receiving is related to the Precision or attention on the lower level perceptual task so from the point of view of this orange agent its generative process its Niche is the blue

agent the blue agent Niche could be considered to be like the out side world but that's where you actually get to plugging

in essentially systems across the interface everything about the generative model like defining the agents generative model is is defining your your um race car and then you have the

track so you can generate a a really sophisticated model but essentially it's just going to stand freely until you close the loop with something even if it's trivial like the environment just um hands out random

observations and it just soaks up and absorbs the actions the actions have no efficacy you could do that then then the agent's generative model policy selection doesn't matter

right so the the context within a dream often just gets like rapidly moved around so the environment of this like internal representational world is kind

of always changing but then if you're in
the process of waking up some looks like
it's growing you know back into Baseline
reality you like slowly P open up
different doors of perception
right h I I I like the doors of
perception
uh connection and also makes me think
like when things are quote normal
seeming it's like me looking at my desk
and and uh you know seeing a little
flashlight it's like well that didn't
take inspiration
because the flashlights actually there
and then it's like but then if it were
something Fantastical then that but then
it's then it's kind of like but if it
was Fantastical and
uninvited then that's a very fine line
with like is that inspiration
or is it just
delusional if it's something that's
that's not that's not that others don't
see on the
desk it's like okay inspiration but then
is inspiration always associated with a
positive
sentiment like motivation or like with
the veillance or with a feeling like
some something has been achieved or
could be achieved
or you know is is in is the is our
definition of inspiration going to
include like a
persistent um anxiety
inducing delusion
yeah it's a good point like a
hallucination we're not like converging

on reality I think is what you're saying you guys have done these before I'm assuming I'm not usually you've done this like thought experiment before this model but I'm assuming it's not usually in such a abstract dream space concept applying active inference models in different

settings like yes yeah yes definitely all all paste in the chat like if you want to see here several dozen code examples um these these um are written by many different

authors and then what the textbook since 2022 and what this other um collaborative work has done has like helped us set up more of that recipe and and the the procedures around it so that like within one to few

conversations across these different settings for cognitive

modeling like we can actually move forward with a domain specific application which is kind of the goal of the textbook it just the textbook

doesn't take one all the way there but that's that's the kernel and that's helping familiarize with the structure of the model capacities some kind of features and uh aspects of the model and then anytime you apply it into a domain this is pretty much what it comes to look

like you just okay here's the general model

schema be like if we were doing a linear aggression on Healthcare be like okay observations are the healthcare data

hidden state is the risk odds for this
or that let's do the
regression this is like that except
there's this downstairs sense making
component and the upstairs action
selection
component but yes we did CH choose a
very
um with time and then with this like
with all these unknown unknowns where
where are What is the niche is is it an
is the
dreamer creating a a a mental Niche for
themselves or
so some some useful elements but with
some things that are simpler in if this
had been a um robot moving on a
chessboard it would have been a simpler
model appreciate there this context it's
cool yeah it's fun it is fun and and
here's here's uh all here's uh
yans he did this for his internship in
The Institute but he used those those
figures from lars's
paper so this was just about time
proximate perception sensory perception
precision
and then attention on
time yielding a metacognitive estimate
of how much time has
passed it's really cool yeah it is
really
cool and then that's kind of again where
the interoperability of these models can
be can be really powerful let's just say
like this is a very strong time
perception or you know let's just say we
have this and other time perception

kernels then somebody on the music and
aesthetic side has made a music
then there's at least a path towards
fusing the time and the Aesthetics
model without needing
to perhaps go to square zero for both of
them I'll copy that into the
thank you for that it also with the uh L
paper that you were talking about
earlier kind of makes me it leads into a
deeper question that I have because I
just don't feel like I have a solid
understanding of how time is put into
these models and the relativity thing
makes me question how space is put into
these models it makes me feel like
there's going to be some kind of curis
some kind kind of motif that models are
generally going to have in order to move
through their environments that we
relate to what we perceive to be time or
what we perceive to be space and so I
think that there's going to be some
relationship of variables that goes to
that but whenever it comes to the hard
definition of what is time whenever
we're talking about this and I might be
overthinking it but I think that there's
still an open question
there
yeah I think I think so like for for
um agents that work in time but don't
necessarily have time awareness like if
there was going to be a 100
robots and they were just unfolding in
time but they didn't necessarily have to
like bootstrap their own time
concept that would be a simpler

setting but then when when we start to
kind of take the Chris Fields physics as
information processing seriously
and if we want to get to the first
principles then we're dealing with
agents constructing their own spatial
temporal reference frames and then as
the modeler there's that question of
like how how does the modeler or the
model sew that fabric
together like if if two agents have
different um one one thinks it's 2
o'clock one thinks it's five
o'clock but then like
is the is the modeler GNA just say okay
the earlier one is right or we flip a
coin to decide who's right it's like
will those will they even be able to
talk to each other and then if the
modeler says well yeah it's 3 o' they're
both
wrong then
then then what system are you
modeling if you're modeling how time
delusion can can um can coexist then
that's one versus I don't know
it it really comes down to what you're
trying to model and and then more of the
the research and speculation side might
take us to what structure could the
models have and then more on like the RX
and for RX environment side we'll
Explore More empirically like what are
the nuts and bolts of
getting of um different
time agent agent specific time systems
to to
interact but that's that here it's this

is just sort of the

skeleton

well

okay facilitators time question

mark um in the coming weeks will go to

chapter eight continuous time so that'll

actually be fun and uh exit the discreet

into continuous time

setting um chapter nine putting

empirical data in rather than just

generating synthetic data fitting with

empirical data I hope by that point

whether with pmdp or another method

let's let's like get an example for that

10 a wrapup or if you are going to be

coming to the chapter 7 then we have

chapter three coming up the high road

chapter four generative models and then

chapter five neurobiology so

cool thank you

all thank you so much cool see you next

time bye